Computer Vision and Machine Learning for Mineral Detection and Classification

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Abstract

This project examines the feasibility of applying computer vision and artificial intelligence techniques for the visual detection and classification of minerals—specifically oxides, sulfides, and silica—based on images acquired under real-world operational conditions. The proposed approach does not aim to replace conventional mineralogical analysis methods; instead, it investigates the potential of deep learning models as auxiliary tools capable of delivering preliminary visual insights that complement established technical procedures. To this end, a rigorous methodological workflow is proposed, encompassing real data acquisition, collaborative dataset curation, expert-driven data labeling, supervised model training, and a critical evaluation of the model's performance and limitations.

1 Introduction

The mining industry is undergoing rapid digitalization [1], and the identification and classification of minerals constitute a fundamental process across multiple stages of mining operations. Traditionally, these tasks rely on physicochemical analyses and expert visual inspection which, although highly accurate, are often costly and time-consuming. In this context, the present project explores the use of computer vision and deep learning models as supportive tools for the preliminary visual classification of minerals based on images captured under real operating conditions.

2 Methodology

The system workflow is organized into five main stages, ranging from the acquisition of visual data to the application of the trained model for inference tasks.

2.1 Visual Data Acquisition

Video recordings captured using fixed cameras are employed. These recordings represent real operating conditions in terms of lighting, resolution, and material variability.

Figure 1: Conveyor belt system used for mineral transportation and data acquisition.

2.2 Dataset Preparation

From the acquired videos, representative frames are extracted through automated processes. The resulting images are standardized in terms of format and resolution and subsequently organized into batches.

2.3 Collaborative Annotation and Curation

The images are uploaded to a collaborative annotation platform or processed individually for later integration. At this stage, regions of interest are delineated using bounding boxes, and the corresponding classes are assigned. As an initial aid to accelerate the annotation process, general-purpose segmentation models such as **SAM/SAM2/SAM3** [2] are employed; however, human review and validation are always maintained to ensure coherence, consistency, and adherence to mineralogical criteria.

2.4 Model Training

The curated dataset is exported in a format compatible with **YOLO** [3] and used to train an object detection model under a supervised learning scheme. The process includes splitting the data into training and validation sets, as well as evaluating model performance using standard metrics.

2.5 Use and Inference

The trained model can be applied to new images or video frames to identify and visually localize the previously defined mineral types.

3 Proposal

The process begins (1) with data acquisition through the capture of video recordings of the material using fixed cameras installed under real operating conditions, for example, above a conveyor belt. (2) Subsequently, visual data extraction and preprocessing are carried out, in which the videos are decomposed into frames using tools such as FFmpeg, and Python scripts are applied to automate selection, filtering, and preparation tasks, thus, standardizing image resolution and format to ensure dataset consistency. (3) The preprocessed images are then managed by uploading them to an annotation and dataset management platform, where they are organized in order to do the labeling and curation stage,(4) regions of interest are delineated using bounding boxes, with support from segmentation-assisted models such as SAM to accelerate the process, while final validation is performed by human experts following mineralogical criteria to ensure annotation quality and consistency.(5) The curated dataset is subsequently exported to a YOLO-compatible format and used for supervised model training implemented in PyTorch, including the separation of training and validation datasets and performance evaluation using standard metrics.(6) Finally, the trained model is applied during the inference stage to new images or video streams to perform visual detection, localization, and classification of minerals, generating results that support operational decision-making.

4 Conclusion

As a final outcome, the project will deliver a trained and evaluated computer vision model, a fully labeled dataset, and a technical analysis addressing performance, limitations, and opportunities for improvement.

Furthermore, this work aims to serve as a foundation for continued development across additional stages of the mineral transportation process, with a particular focus on fault prevention in critical equipment such as primary and secondary crushers. By extending the proposed framework, the system seeks to contribute to improved operational reliability and proactive maintenance strategies within mining operations.

References

[1] A. Soofastaei (Ed.), *Advanced Analytics in Mining Engineering: Leverage Advanced Analytics in Mining Industry to Make Better Business Decisions*, Springer, 2022.

[2] *SAM 3: Segment Anything with Concepts.* Available at: https://docs.ultralytics.com/models/sam-3/

[3] *Models Supported by Ultralytics.* Available at: https://docs.ultralytics.com/models/